

Czech News Corpus Search

Full-text search and retrieval pipeline over 1,000 Czech news articles.

THE ASSIGNMENT

NLP task

Goals:

1. Solution design

Design and implement a basic **retrieval pipeline** for searching in a corpus of news articles.

- Build a **full-text search** mechanism, including relevant NLP techniques.
- Focus on high-quality search and retrieval functionality.
- Design and describe the **evaluation methodology**. How would you assess the system's effectiveness?
- Reflect on the **limitations** of your current implementation and suggest **potential improvements**.

2. Industrialization

Draw an **architecture diagram** how the solution can be designed in **production**.

02 • The Problem



THE CHALLENGES OF THIS SPECIFIC CASE

Czech morphology

Praha = Prahy = Praze • 7 cases

No ground truth

zero manual labels available

Length variance

431 to 20,000 characters

A PROBLEM RELEVANT FOR DATASENTICS CLIENTS

Czech bank • Compliance & Risk

- ▶ 10,000+ internal policy documents, never leaving the bank's infrastructure
- ▶ Analysts query in natural language
- ▶ Fully self-hosted pipeline: no data sent to external APIs

03 · Agenda

1

Context

Dataset · State of
the art of retrieval
methods



2

Pipeline

My implementation of
retrieval methods



3

Evaluation & Results

Ground truth · 3
protocols compared



4

Production

FastAPI · Docker



5

Takeaways

What I learned ·
Limitations

04 · 1,000 Czech news articles · simecek/czech_news

headline

Hamáček oznámil
předsednictvu ČSSD, že
končí ve funkci předsedy

*Jan Hamáček announced
his resignation as ČSSD
party chairman*

brief

Dosluhující vicepremiér a
ministr vnitra Jan Hamáček
oznámil...

*Outgoing deputy PM Jan
Hamáček announced...*

content

431 to 20,000 characters

keywords

Jan hamáček · Rezignace ·
Předseda · Čssd · Volby

Domácí

KEY INSIGHT

- 46× length ratio between shortest and longest article
- High variance forces a chunking strategy
- A single embedding cannot represent a 20,000-character article faithfully

05 · State of the art of retrieval methods

Sparse BM25

Exact term matching · no GPU · fast

Query



Tokenize · Score



Ranked articles

Dense embeddings

Semantic similarity · handles morphology

Query



Embed · Cosine similarity



Ranked articles

Hybrid RRF fusion

Parameter-free rank fusion · no score normalization needed

Query



BM25 results + Dense results · Rank fusion



Ranked articles

06 · My implementation of retrieval methods

Sparse BM25

FIELD WEIGHTING

headline ×2 + **brief** ×2 + **content** ×1

BM25Okapi · tokenized via `.split()`

EXAMPLE

"Hamáček" appears twice

Headline + brief → counted twice → ranks higher

WHY BM25Okapi?

- ▶ Available in rank-bm25, no heavy dependencies
- ▶ No labeled data to train a learned ranker
- ▶ BM25L / BM25+ not empirically tested, assumed limited benefit at this corpus size

RUNTIME BM25 indexing ~2.3 s

Dense embeddings

paraphrase-multilingual-MiniLM-L12-v2

384 dim · 12 layers · no GPU

50+ languages incl. Czech

CHUNKING

2,000 chars / chunk

200 char overlap

2,117 chunks / 1,000 art.

Search by chunk, return full article

WHY MiniLM-L12?

- ▶ CPU-only, lightweight, 50+ languages including Czech
- ▶ multilingual-e5-large is higher quality but requires a GPU, too heavy for a full-stack assignment

RUNTIME Embedding gen (CPU) ~3 min 54 s

Hybrid RRF fusion

BM25 top-100 + Dense top-100

• RRF **k=60**

WHY RRF?

- ▶ Parameter-free, only k=60, standard default from the original RRF paper
- ▶ Weighted blending requires normalizing BM25 and cosine scores that live on incomparable scales, brittle
- ▶ RRF uses rank position only, not raw score, no normalization needed

07 • Ground Truth • 3 protocols

Methodology: keyword weak supervision → LLM-generated queries → lexically-independent semantic queries

KEYWORD GT	SYNTHETIC GT	SEMANTIC GT
SOURCE Article keywords from metadata STRATEGY 1 • precision All keywords of one article as query → expects that article at rank 1 STRATEGY 2 • recall Single keyword as query → expects every tagged article in top 10 BIAS Favors BM25 by construction	SOURCE Claude API generated queries QUERY STYLE Natural language, topic based BIAS May reflect LLM patterns	SOURCE Claude API, zero lexical overlap enforced QUERY STYLE Paraphrase style, no shared words BIAS Zero lexical overlap enforced, no word matching shortcut possible

50 queries per protocol, ground truth = source article, metric = P@1 and MRR@10

08 • Results • 3 protocols compared

METRIC	Sparse (BM25)	Dense	Hybrid (RRF)
Keyword GT · P@1	0.62	0.25	0.43
Keyword GT · MRR@10	0.69	0.32	0.53
Keyword GT · Recall@10	0.41	0.19	0.35
Synthetic GT · P@1	0.74	0.64	0.74
Synthetic GT · MRR@10	0.80	0.69	0.82
Semantic GT · P@1	0.44	0.48	0.48
Semantic GT · MRR@10	0.53	0.59	0.58

150 queries • 50 per protocol • Claude API claude-sonnet-4-6

KEY INSIGHT

Hybrid is the most stable performer across all three protocols, though the margins stay small and aren't statistically validated, which is exactly why it's the safer default in production across query types.

CAVEAT • IN PRODUCTION

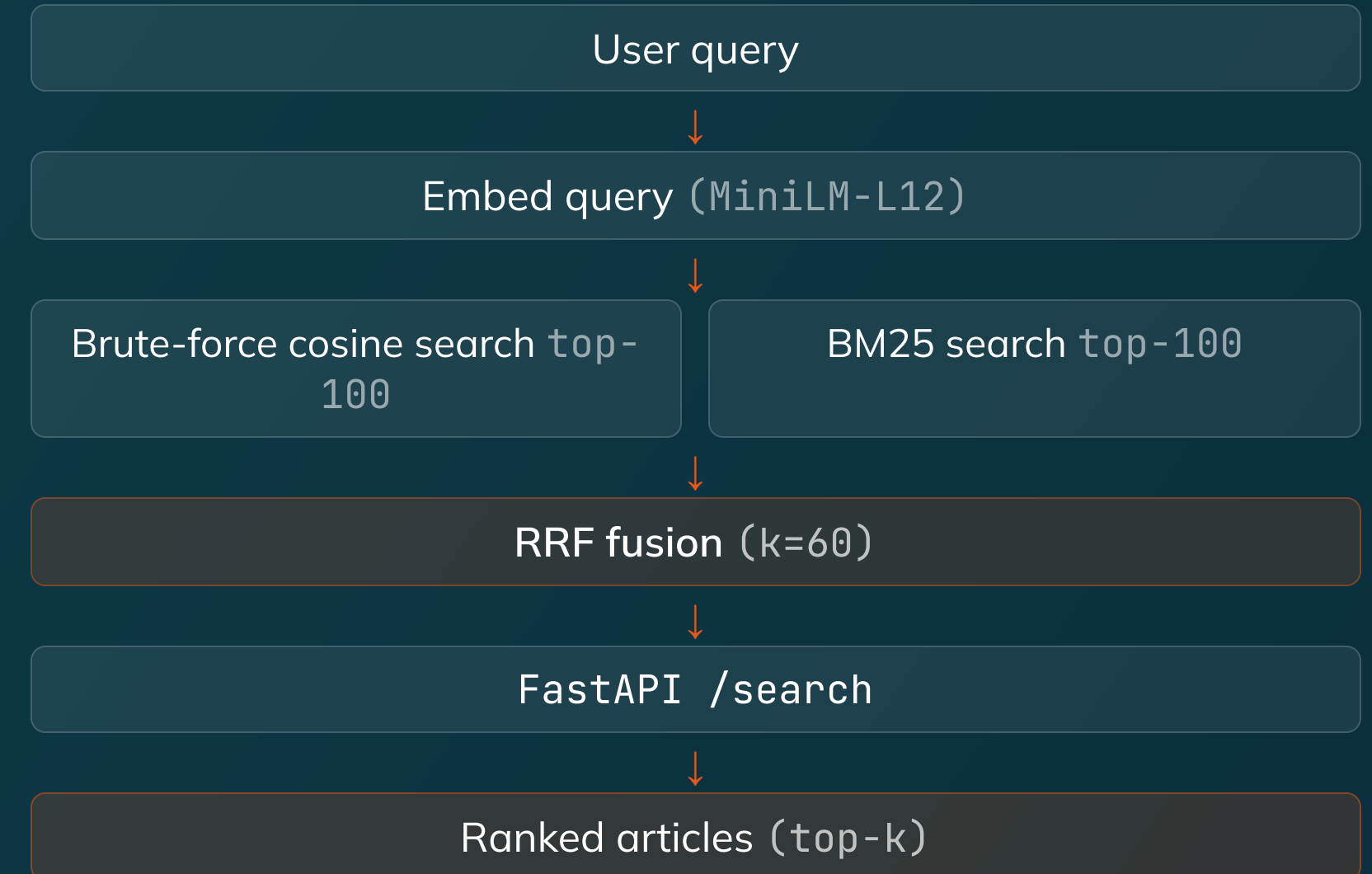
50 queries per protocol limits statistical confidence (no bootstrap CIs yet).

9 · Architecture production

INGESTION · OFFLINE



QUERY PATH · REAL-TIME



RUNTIME Query latency ~62 ms

Brute-force cosine

Sufficient at 2,117 chunks. Milvus becomes necessary at scale, see Scale in slide 10.

FastAPI + Uvicorn

always-on REST server

Docker

Containerized deployment, port 8000, embeddings mounted as volume

10 • Limitations • What's next

LIMITATIONS

Evaluation

- ▶ 50 queries per protocol, no bootstrap CIs, gaps likely not significant
- ▶ No manual relevance judgments, weak supervision and synthetic labels only, priority fix: 100–200 annotated queries
- ▶ Claude-generated queries (Synthetic and Semantic GT) may reflect LLM patterns, cloud US API vs sovereign-deployment tension

Retrieval quality

- ▶ Field weighting (headline×2, brief×2), chunk size (2,000/200), and RRF k=60 are reasoned defaults, not grid-searched on this corpus
- ▶ BM25L / BM25+ assumed beneficial only by reasoning, never empirically tested here
- ▶ Semantic GT P@1 = 0.48, not production-ready without precision gains

Scale & hardening

- ▶ Brute-force cosine breaks at 100M articles
- ▶ Motivates Milvus + ANN
- ▶ No authentication, rate limiting, or structured logging on the FastAPI endpoint yet

WHAT'S NEXT

EVALUATION

- ▶ **Human-labeled benchmark:** 100–200 queries
- ▶ **Statistical validation:** bootstrap CIs
- ▶ **Self-hosted open source LLM for query generation:** removes Claude API sovereignty tension

RETRIEVAL QUALITY

- ▶ **Czech lemmatization:** cut morphological noise for BM25
- ▶ **bge-reranker-v2-m3:** cross-encoder on top-20, self-hosted
- ▶ **BM25L / BM25+:** correct BM25Okapi normalization bias on longer documents
- ▶ **Performance analysis by article length:** does retrieval quality degrade on 20,000-character articles?

SCALE & HARDENING

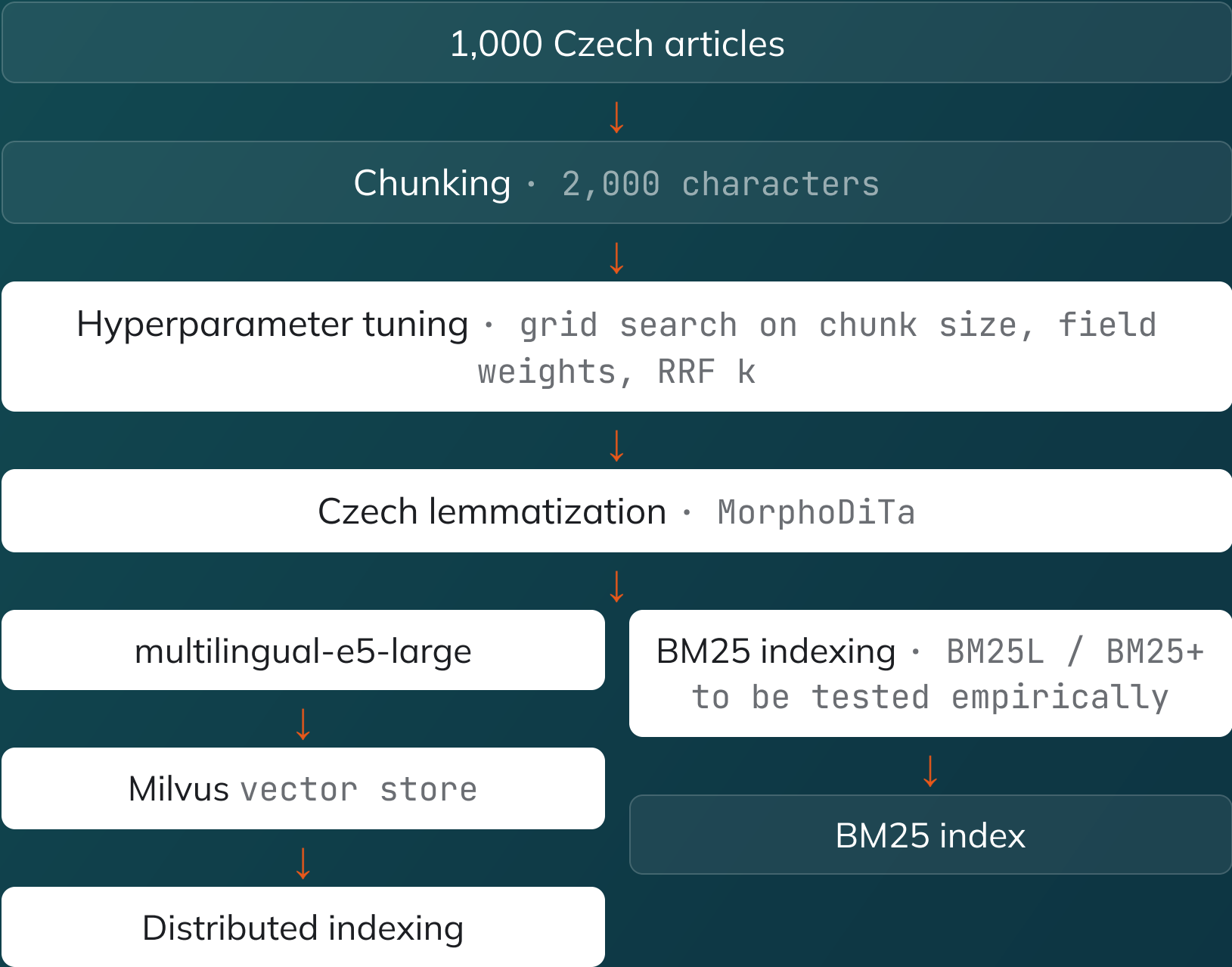
- ▶ **Milvus + ANN:** replace brute-force cosine
- ▶ **Distributed indexing:** handle corpus growth
- ▶ **API hardening:** auth, rate limiting, structured logging

PRODUCTION MONITORING

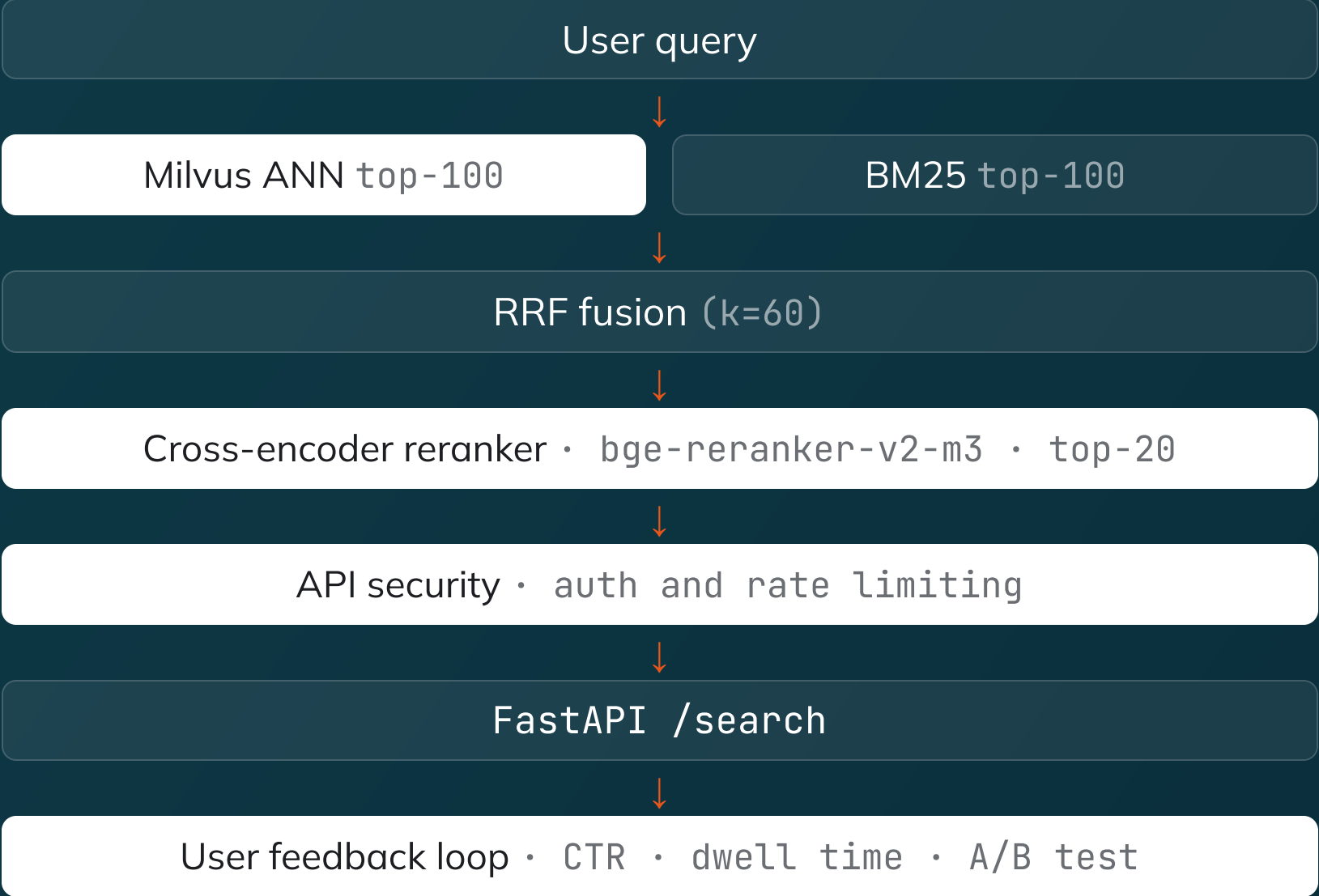
- ▶ **Online signals:** CTR, click position, dwell time
- ▶ **A/B testing:** compare versions on real traffic
- ▶ **Feedback loop:** build a human-labeled set over time

11 • What the pipeline looks like with improvements

INGESTION • OFFLINE



QUERY PATH • REAL-TIME



► bge-reranker-v2-m3: open-source, multilingual, self-hosted. Consistent with sovereign deployment approach

■ Dark box = current ■ White box = proposed improvements

Let's discuss

QUESTIONS I'D LOVE YOUR PERSPECTIVE ON

- ▶ How do you evaluate retrieval quality in production without human labels?
- ▶ Do your clients accept synthetic evaluation as a proxy?
- ▶ When do you introduce rerankers in a production pipeline?
- ▶ How important is model sovereignty when selecting embedding models?

WHAT THIS PROJECT SHOWED ME

- ▶ Evaluation design is as important as retrieval design
- ▶ Engineering pragmatism: ship the full pipeline first, optimize components second
- ▶ Czech morphology is a real constraint. Lemmatization would be my first retrieval upgrade